Weak points in the ECDLP hardness corpus: triage, 2026-11-03

Question. Where is the claim graph that supports ECDLP hardness weakest, and which weak points can be broken cheaply enough to attack first?

Scope. The assembled corpus (version 2026-11-01: 583 claims, 950 edges, 236 sources, including the protocol-layer port of W7 and the Lean results of 2026-11-04), read through `weakness.mjs`, plus the repository sources the corpus cites. W1–W7 change no existing claim's status; proposed edits are listed as actions. The Lean round of 2026-11-04 adds kernel-checked nodes, refutes one inference (`CLAIM-MARGINAL-IMPLIES-HAZARD`) and moves the xedni waiting-time edge onto the weaker per-trial marginal assumption. The generated companion is `WEAKNESSES.md` (all rows) with `weakness.json` and `weakness-queue.svg`.

Method. `attack-graph.mjs` already ranks unestablished claims by impact (how much rests on them) and propagates taint from refuted claims. `weakness.mjs` adds what that pass does not see:

check	what it finds	count
status inflation	proved / algorithm / measured claims whose support closure reaches an open, heuristic or assumed claim	17
unwired dependence	a claim's own assumption text invokes a heuristic node with no edge to it	11
live obstruction	an established claim `obstructs` / `qualifies` / `does_not_cover` a claim that still supports others	49 (17 `obstructs`)
route closure on weak support	an `obstructs` edge (a closed attack route) whose obstruction rests on an assumed, heuristic or open claim	16
hidden shared assumption	the same assumption text on 3+ nodes, never a node itself	18 clusters
source contagion	a primary source with a refuted claim and other live claims	21 sources
model-scope leap	generic-model or measured claim supporting a broader one without `model_bridge`	9

It then ranks the 246 attackable claims by impact × breakability, where breakability is the expected cost of a decisive result: a heuristic with a stated formula in an algebraic model can be checked exactly on small fields; an open literature conjecture cannot. Established `obstructs` edges multiply the score. Reading a primary text or auditing a kernel-checked statement goes in a separate hygiene queue (100 claims): cheap, but it rarely overturns anything. **The scores are a triage heuristic, not evidence.**

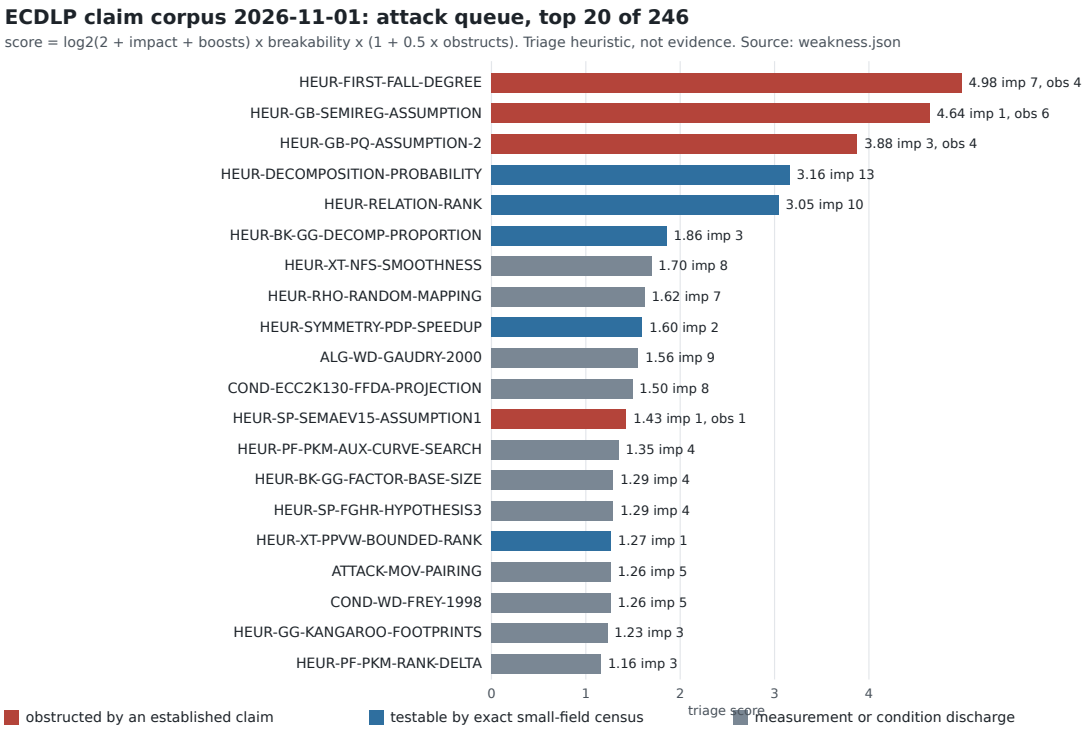


Figure 1: Attack queue, top 20 (weakness-queue.svg)

Evidence labels below: **exact** = a count or graph fact over the corpus files; **read** = taken from a cited repository source, quoted path given; **derived** = an argument written here, not yet reviewed or kernel-checked; **scratch** = a toy check run in this session's scratch space, not committed and not evidence for the corpus.

W1. The first-fall-degree pillar is obstructed and still load-bearing

Exact. `HEUR-FIRST-FALL-DEGREE` has four incoming `obstructs` edges from established claims (CG21 incomparability, Kosters–Yeo's first fall degree 2 on descended S_3 at every n, and two Caminata–Gorla summation measurements), and it still conditions seven claims: `COND-ECC2K130-FFDA-PROJECTION` (the 2^86 projection), `COND-NAGAO-2015-COMPLEXITY`, `COND-PKM-PRIME-ALGEBRAIC`, `COND-BINARY-WEIL-DESCENT`, `COND-GB-HFE-FIRST-FALL-ESTIMATES`, `COND-SP-HPST-SYMMETRIZED-PRIME-DEGREE`, `COND-SP-KARABINA-PDP`. Its siblings are in the same state: `HEUR-GB-SEMIREG-ASSUMPTION` has six `obstructs` edges, `HEUR-GB-PQ-ASSUMPTION-2` four. These three take the top of the attack queue. `HEUR-GB-PQ-ASSUMPTION-2` also states the first-fall assumption in its own text with no edge to `HEUR-FIRST-FALL-DEGREE`.

Judgment. As a general principle the first-fall-degree assumption is already dead in the corpus's own terms. What survives is a family-specific version for each estimate that uses it. The graph keeps it at `heuristic`, so the seven conditional estimates read as one discharged condition away from usable, which overstates them.

Attack. Petit–Quisquater Assumption 2 is asymptotic (`D_reg = D_ff + o(1)`), so no finite table refutes it; the repository's measurements (D_ff = 2, degree 4→5 at n = 36→37, 5 at n = 45) only obstruct it. A **lower-bound proof** settles it: show that the solving degree of the descended S_3 system at m = 2 is ≥ 3 for all $n \geq n_0$. Degrees are integers, so `+ o(1)` means eventual equality, and a uniform lower bound of 3 with D_ff = 2 refutes the assumption for that family. Candidate route: exhibit, for every n, a degree-2 Macaulay span that provably misses a polynomial the reduced Gröbner basis needs. A census of the degree-2 Macaulay rank against n over F_2 would show whether the gap is uniform before the proof is attempted.

Effect on the baseline. None directly. This retracts projected IC costs that were never the measured baseline. It matters because it removes a whole family of "conditional subexponential" rows from the menu of routes worth funding.

W2. The relation-rank heuristic is load-bearing only by a modelling choice

Exact. `HEUR-RELATION-RANK` conditions `ALG-SPARSE-LINEAR-ALGEBRA`, and through it the corpus marks `THM-WD-GTTD-THM1`, `THM-WD-DIEM-SMALL-GENUS` and `ATTACK-GHS-WEIL-DESCENT` (all `proved_external`) as resting on a heuristic: 4 of the 17 status-inflation rows. Five more claims invoke rank in their assumption text with no edge (W5).

Read. `HEUR-BK-GG-RIGHT-KERNEL-RANK` (Galbraith–Gaudry 2016, §6.2) already records that left-kernel (target-dependent) relations "are much easier to analyze rigorously", while right-kernel relations need rank #F − 1. No node carries the left-kernel argument and no edge discharges the rank heuristic with it.

Derived (needs review and a Lean statement). Let r be prime and Q = xP. Draw (a_i, b_i) uniformly, set $R_i = a_iP + b_iQ$, and keep the R_i that decompose over the factor base F, giving rows $M_i \in (Z/r)^{\wedge}F$. Take B = |F| + 1 rows. Some nonzero λ satisfies λM = 0, whatever the rank of M, and then $\Sigma\lambda_i a_i + x \cdot \Sigma\lambda_i b_i \equiv 0 \pmod r$. Fix the points R_i: the decomposition, and so M and λ, depend only on the R_i. Given R_i, b_i is still uniform on Z/r (the r pairs with $a_i + b_i x = \log R_i$ are equally likely). So $\Pr[\Sigma\lambda_i b_i \equiv 0] = 1/r$ and x is recovered with probability $1 - 1/r$. Signed/automorphism-folded columns change nothing. **No rank condition is**

used. That is why GTTD and Diem can be rigorous, and the four inflation rows are edge errors, not fragile theorems.

Scratch. Toy check (p = 10007, prime-order curves, factor base x < 60, two-term decompositions, |F| + 1 relations, 12 seeds): every matrix was rank-deficient (2–7 short of |F|), so the right-kernel formulation fails on every one, and the left-kernel recovery returned the correct log in 12/12. Illustration only.

Where the heuristic is truly load-bearing. Only where relations are reused across targets (right-kernel amortisation): multi-target batches such as the `ic.residual_batch` vs `rho.multitarget_pooled` parity work, and PKM's $\Delta = 10$ surplus (`HEUR-PF-PKM-RANK-DELTA`). There the measurable question is the rank deficit as a function of |F| and the column-folding convention, which a native census can answer exactly.

Actions. (a) Add a `THM-LEFT-KERNEL-RELATIONS` node (derived above; formalise the finite core in Lean: uniformity of b_i given R_i and the 1/r bound). (b) Re-wire the rigorous Weil-descent chain to it, which removes 4 inflation rows. (c) Narrow `HEUR-RELATION-RANK` to the multi-target / right-kernel regime and add the missing edges from W5. (d) Measure rank deficit on the multi-target factor bases the repository already uses.

W3. The decomposition-yield heuristic conflates two regimes

Exact. `HEUR-DECOMPOSITION-PROBABILITY` has the largest impact on the frontier (13), and four more claims invoke it in text without an edge (W5). Its statement already says the mean B^m/r "follows by counting": only the distribution shape (Poisson, variance, independence, rationality, solver visibility) is heuristic.

Judgment. Two different claims share this node. Over F_{q^n} with n fixed, Diem's rigorous results (`THM-WD-DIEM-SMALL-GENUS`; its scope records the rigorous q^(2−2/n) ECDLP bound cited by KMPS) bypass the heuristic. Over prime fields it is genuinely heuristic, and that is where this repository's programme lives. Under the left-kernel formulation (W2), the only quantity an index-calculus cost needs is Pr[a uniform point decomposes] = |{R : R decomposes}| / r, which is a **set size, not a heuristic**, and exact on small fields.

Attack. Split the node into an extension-field version (cite Diem; status follows the read) and a prime-field version. For the prime-field one, run an exact census over every target on a scaling family of small p. Record the full histogram of representation counts, not just the mean, and report the observed decomposable fraction against $1 - \exp(-\mu)$ with its trend in p. A systematic deviation is a

correction to every prime-field yield estimate; agreement is a measured bound. Either outcome is decisive at the measured scale.

W4. The Lemma 4 refutation rests on a reading choice, and two internal papers disagree

Exact. Eleven nodes in the Nagao Lemma 4 family carry the free-text assumption "Equality reading of Definitions 5 and 6", the largest hidden-assumption cluster. `CLAIM-NAGAO-LEMMA4` is `refuted`, and its scope says the refutation is reading-specific.

Read. `papers/ecdlp-boundaries-20261018/sections/fall-degree.tex` tabulates all four (reading × placement) cells. The equality reading is refuted both outside and inside; literal ≥ outside is refuted (d_F = 3, d'_F = 2); literal ≥ inside gives d_F = d'_F = 2. It also states that the wording "the literal reading always voids the counterexample is too broad". But `papers/ic-negative-results/ic_negative_results.tex` (≈ line 470), the repository source for the refutation (`S-IC-NEGATIVE`), still says "under that literal reading the lemma holds only vacuously". The boundaries section also notes that a final publication "still owes a fresh PDF-level transcription of both definitions".

Actions. (a) Correct the ic-negative wording to the boundaries table. (b) Record the reading as a parameter of `CLAIM-NAGAO-LEMMA4` (refuted in 3 of 4 cells) instead of a footnote. (c) Run the existing exact census on the one open cell (literal ≥, field equations inside). (d) Do the PDF transcription, because the refutation is only as good as the reading.

W5. Eleven unwired dependences understate the top frontier nodes

Exact. These claims name a heuristic in their own assumptions but have no support path to its node, so refuting the heuristic would not reach them:

- decomposition yield → `ALG-GAUDRY-FIXED-EXTENSION`, `COND-BINARY-WEIL-DESCENT`, `COND-GB-PQ-SUBEXP-BINARY-ECDLP`, `COND-SP-KARABINA-PDP`
- relation rank → `COND-PF-PKM-COST-THRESHOLD`, `ALG-GAUDRY-FIXED-EXTENSION`, `COND-BINARY-WEIL-DESCENT`, `COND-GB-PQ-SUBEXP-BINARY-ECDLP`, `COND-PKM-PRIME-ALGEBRAIC`, `HEUR-XT-NFS-SMOOTHNESS` (NFS rank is the same phenomenon in a finite-field setting; wire it or record why not)
- first fall degree → `HEUR-GB-PQ-ASSUMPTION-2`

Action. Add the `conditions` edges, then re-run `attack-graph.mjs`. `weakness.mjs` now reports any new occurrence.

W6. Smaller structural findings

- **Conditional obstruction labelled proved (exact).** `OBSTRUCTION-XEDNI-HEIGHT` is `proved_external`, but it rests on Lang's height conjecture (`CONJ-XT-LANG-HEIGHT`, open) and six more hypotheses its own scope lists. The xedni line is closed only conditionally. Relabel it `conditional`. As an attack route it is still poor (it needs a uniform failure of Lang's conjecture on the selected lifts).
- **In-model theorems resting on an assumption (exact).** Five rho theorems (KMPT, KMT, MV mixing and nondegeneracy, three-partition) are `proved_external` given `ASM-GG-RANDOM-PARTITION`. That is correct inside the model, but implemented r-adding walks are not random partitions. This affects the **rho reference**, not IC: the repository already matches rho empirically, which is the right control.
- **Source contagion (exact).** Among primary sources, Semaev 2015 has 2 of its claims refuted while `HEUR-SP-SEMAEV15-ASSUMPTION1` ($d_{F4} \leq 4$) is still live and already obstructed once (Kosters–Yeo: degree 5 at $n = 45$). `RT-SP-SEMAEV-A1-N45` is the decisive test and is cheap. Kosters–Yeo, Euler–Petit/HKPYY and BGJT each have one refuted claim and live siblings (listed in `WEAKNESSES.md` §4).
- **Model-scope leaps (exact).** All nine are generic-model rho heuristics feeding standards or the concrete rho algorithm. That is the honest, well-known generic-to-concrete step, and the repository's matched-rho measurements control it.

W7. The protocol-layer corpus, ported and cross-linked

What. `protocol-attack-corpus.json` ports the sibling repository's cryptographic assumption attack corpus (crypto-autoresearcher `research/cryptographic_assumption_attack_corpus_20261101/`, pinned at commit `00435557`) into this schema: 37 nodes, 34 edges, 2 research targets.

- **The finished five.** Each becomes a refuted overbroad claim plus the counterexample or repair that refutes it, carrying the sibling's Lean theorem names (all five passed the sibling's semantic review). Their Lean files live in the sibling, so no `formal` block is recorded here, and their status stays `proved_internal_unreviewed` for this corpus.
- **The queue.** The Priority 0–2 queue becomes 23 open obligations.

Read (cross-links). The sibling's Lean results now attach to the ECDLP claims whose own assumption text they discharge or qualify:

- `RED-SCHNORR-ROM` lists "Correct subgroup validation" and "Reduction obtains two related accepting transcripts"; the SUBGROUP countermodel qualifies the first and the Schnorr special-soundness theorem enables the second.

- SUBGROUP also qualifies BOUNDARY-INVALID-CURVE and THEOREM-RANDOM-SELF-REDUCTION.
- The queue items qualify LB-SHOUP-DLP (transparent encodings), the AGM transfer claims, HEUR-RHO-RANDOM-MAPPING, STANDARD-CURVE-EXCLUSIONS, the cofactor and CDH nodes.
- POLY-FUNCTION qualifies CLAIM-NAGAO-LEMMA4. The true/fake fall-degree gap is the formal-versus-function gap: $X_j^2 + X_j$ vanishes as a function on F_2 but not formally, which is exactly why reducing modulo the field equations changes the degree.

Promoted assumption. OBSTRUCTION-XEDNI-HEIGHT listed "a history-uniform conditional-hazard recurrence for adaptive trials" as free text. It is now the node ASM-XEDNI-HAZARD-RECURRENCE (assumed), conditioning the obstruction. The protocol corpus's ONE-SHOT-REPEAT item qualifies it: a one-shot bound alone does not give a geometric waiting time under correlated or adaptive trials. RT-PA-XEDNI-HAZARD asks for either the recurrence for the JKSST selected-lift process or an adaptive countermodel.

Exact (new lens: route closures on weak support). Impact counts what rests on a claim, so it cannot see the value of breaking an obstruction, which usually supports nothing downstream. Breaking an obstruction's support **reopens an attack route**, which is the direction that matters for beating rho. weakness.mjs now lists every obstructs edge whose obstruction rests on an assumed, heuristic or open claim: 16. Read by route:

- **Xedni is closed twice, both times softly.** HEUR-XT-EASY-LIFT-DEPENDENCE (heuristic, Silverman's Masser-based argument), and OBSTRUCTION-XEDNI-HEIGHT, which rests on Lang's conjecture and the hazard recurrence.
- **Lifted index calculus is closed by three heuristics.** Miller's height argument, PPVW bounded rank, and the Silverman–Suzuki analysis.
- **The naive prime-field Semaev method is closed** by YYTK's lower bound, conditional on their statistical Conjecture 10, which was validated only up to about 25-bit primes. The closure's scope already excludes PKM towers, APS partitions and non-Gröbner solvers.
- **Quasi-subfield-polynomial routes are closed** by "no gain" results resting on HEUR-DECOMPOSITION-PROBABILITY, the yield heuristic of W3.
- **The prime-field relation search** (GAP-PF-PRIME-RELATION-SEARCH) is closed by HEUR-PF-NO-SUBFIELD-OBSTRUCTION, labelled heuristic.
- **Rank-only closures are not reopenings.** The GHS (KMPS) and two-summand closures rest on HEUR-RELATION-RANK; per W2, removing that heuristic strengthens them.

Derived (strengthens a closure, so it saves effort). One tempting escape from HEUR-PF-NO-SUBFIELD-OBSTRUCTION fails. Base-change E/F_p to F_{p^n}, decompose a target R in E(F_p) over points whose x-coordinates lie in an F_p-subspace V of F_{p^n}, and pull the relation back with the trace: nR = Σ Tr(Pᵢ). The pulled-back factor base Tr(F') has about p^{dim V} ≥ p elements, the size of the whole group, while rho needs about √p. Any subspace-based factor base therefore fails on prime fields by a dimension count, not by heuristic. A prime-field escape needs a factor base that is not a subspace image (small integers, composed rational maps, random sets), which is where the corpus already places PKM and APS.

Attack. Order the reopenings by how cheaply each support can be tested:

1. YYTK Conjecture 10, an exact statistic on summation polynomials, re-measured well past 25 bits.
2. The hazard recurrence for adaptive lift selection: a Lean separation first, then the JKSST process.
3. The easy-lift dependence probability: an exact count on small families against the claimed below-1/p rate.

The three lifting heuristics over Q are deep (height and rank conjectures) and are not cheap targets.

Lean results, 2026-11-04

Lean project `research/ecdlp_weakness_lean_20261104/lean` (Lean v4.34.1, Mathlib v4.34.1). All ten theorems depend only on `propext`, `Classical.choice` and `Quot.sound` (`AxiomAudit.lean`). The repository proof gate re-checks them in CI.

W2 is kernel-checked: `THM-LEFT-KERNEL-RELATIONS`.

- **Existence (`exists_left_kernel`).** More relations than factor-base columns give a nonzero left-kernel vector for every relation matrix. No rank hypothesis is used.
- **Recovery (`left_kernel_recovers`).** That vector recovers the logarithm whenever its b-weight is nonzero.
- **Exact failure count (`card_bad_pairs`).** If the vector is any nonzero-valued function of the relation points alone, exactly q^(2n−1) of the q^(2n) hidden pairs (a, b) fail: density exactly 1/q.

HEUR-RELATION-RANK, HEUR-BK-GG-RIGHT-KERNEL-RANK and the "enough independent relations" assumption of ALG-SPARSE-LINEAR-ALGEBRA are now qualified by it. The corpus still lists the old support edge from the rank heuristic into ALG-SPARSE-

LINEAR-ALGEBRA. Re-wiring the rigorous Weil-descent chain (removing 4 inflation rows) is a separate, reviewable edit.

W7 is kernel-checked: the xedni hazard recurrence is unnecessary.

- **Repair (THM-HAZARD-MARGINAL-SUFFICES).** If each trial succeeds with marginal probability at most ε, then E[min τ N] ≥ N/2 whenever Nε ≤ ½. This is the JKSST audit's waiting-time conclusion without its conditional-hazard recurrence.
- **Separation (CEX-HAZARD-SEPARATION).** A three-outcome model has every marginal ≤ 1/3, yet the recurrence fails (4/9 > 1/3), so CLAIM-MARGINAL-IMPLIES-HAZARD is refuted.
- **Rewiring.** OBSTRUCTION-XEDNI-HEIGHT now rests on the weaker ASM-XEDNI-TRIAL-MARGINAL: each adaptive trial, as actually run, succeeds with marginal probability O(1/p).

This **strengthens** the xedni closure rather than reopening it. The remaining bridge is whether adaptive lift selection can raise a single trial's marginal above the one-shot bound.

Recommended order of attack

#	target	kind of result	cost	what changes if it lands
1	W2: re-wire the rigorous Weil-descent chain to THM-LEFT-KERNEL-RELATIONS (Lean core done 2026-11-04)	corpus edit	low	4 inflation rows resolved; rank heuristic confined to the multi-target regime
2	W5: add the 11 missing edges	corpus fix	trivial	true impact of the top frontier nodes becomes visible
3	W4: fix the ic-negative wording; census the open cell	correction + exact census	low	the Lemma 4 refutation becomes reading-complete
4	W3: prime-field yield census, full histogram, scaling in p	exact measurement	low–medium	every prime-field yield estimate is checked or corrected
5	W1: degree-2 Macaulay rank census, then the lower-bound proof	census, then proof	medium	PQ Assumption 2 refuted for m = 2; the FFDA rows retire
6	RT-SP-SEMAEV-A1-N45	exact F4 run	low	Semaev 2015 Assumption 1 settled at the measured scale
7	W7: YYTK Conjecture 10 past 25 bits	exact census	low–medium	the naive prime-field Semaev closure is confirmed, or the route reopens

8	W7: per-trial marginal bound for adaptive xedni lift selection (`ASM-XEDNI-TRIAL-MARGINAL`; the separation and repair are done in Lean)	proof or countermodel	medium	the xedni waiting-time step is closed, or adaptive selection reopens the route

Items 1, 2, 3 and 8 are corpus and proof work, not experiment runs. Items 4–7 are runs and go through the repository's experiment route (protocol, resume checkpoint, native code, validation gate) before any number enters the corpus.

Reproduce

```
node docs/ecdlp-hardness-corpus/weakness.mjs          # writes WEAKNESSES.md,
weakness.json, weakness-queue.svg
node docs/ecdlp-hardness-corpus/weakness.mjs --check  # fails if they are stale
```